6.5 Properties of Logarithms

Name _____ Period ____ Date _____

Algebra II 6.5 · Student sheet

I can…

- use the properties of logarithms to evaluate logarithms.
- expand and condense logarithmic expressions.
- use the change-of-base formula.

Vocabulary · fill in the missing word

Product Property — $\log_b mn =$ _____

Quotient Property — $\log_b \left(\frac{m}{n}\right) =$ _____

Power Property — $\log_b m^n =$ _____

Change-of-base formula — $\log_c a =$ _____ $= \ln a / \ln c$

Part 1 · Calculator test

Use a calculator. Round to three decimal places. Are the two columns equal?

Expression	Value	Compare with	Value	Equal?
$\log 2 + \log 3$		$\log 6$		
$\log 12 - \log 4$		$\log 3$		
$3 \log 2$		$\log 8$		
$\log 5 + \log 5$		$\log 10$		
$\log 10 - \log 2$		$\log 8$		

Rules: $\log m + \log n =$ _____ $\log m - \log n =$ _____ $n \log m =$ _____

Part 2 · Notes and examples (fill in each note, then work the example)

Product Property: $\log_b mn =$ _____

Quotient Property: $\log_b (m/n) =$ _____

Power Property: $\log_b m^n =$ _____

Example 1

Use $\log_2 3 \approx 1.585$ and $\log_2 7 \approx 2.807$ to evaluate each logarithm.

a. $\log_2 21$ b. $\log_2 \left(\frac{7}{3}\right)$ c. $\log_2 49$

Example 2

Expand $\ln \left(\frac{5x^7}{y}\right)$.

Example 3

Expand $\log_3 (x^2\sqrt{y})$.

Example 4

Condense $\log 9 + 3 \log 2 - \log 3$.

Change of base: $\log_c a =$ _____ = _____

Example 5

Evaluate $\log_3 8$ using common logarithms and natural logarithms.

Example 6

The loudness of a sound is $L(I) = 10 \log \left(\frac{I}{I_0}\right)$, where I is its intensity. How much louder (in decibels) is a sound when its intensity is doubled?

Part 3 · Partner practice

Partner 1

Expand $\log_6 (3x^4)$.

Partner 2

Condense $2 \ln x + \ln 5 - \ln y$.